JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

AUGUST 2025 EXAMINATIONS

OPTION – A011

PHYSICS **Time Allowed: 3 Hours**

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer all questions in this section.

Write the OPTION NUMBER that appears on your question paper on all your answer sheets and booklets

Write your answers legibly in the Answer Sheet provided for Multiple-Choice Questions

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in all; ONE from each Course.

Write the OPTION NUMBER that appears on your question paper on all your answer sheets and booklets

Ensure you read and follow all the Instructions on the cover page of the Answer Booklet.

Turn Over

The following constants are given

Acceleration due to gravity, g	$9.8\ m/s^2$
Atomic mass unit, u	$931.5\ MeV = 1.661 \times 10^{-27}\ kg$
Avogadro's number, N_A	$6.02 \times 10^{23}\ mol^{-1}$
Boltzmann constant, k	$1.38 \times 10^{-23}\ J/K$
Density of water, ρ_w	$1000\ kg/m^3$
e/m	$1.76 \times 10^{11}\ C\ kg^{-1}$
Electric Charge, e	$1.6 \times 10^{-19}\ C$
Electron rest Mass, m_e	$9.11 \times 10^{-31}\ Kg$
Electron Volt, eV	$1.602 \times 10^{-19}\ J$
Gravitational constant, G	$6.67 \times 10^{-11}\ Nm^2/kg^2$
Mass of Helium nucleus	4.0015 u
Molar gas constant, R	8.31 J/K/mol
Permeability of a free space (μ_0)	$4\pi \times 10^{-7}\ Hm^{-1}$
Mass neutron	1.0087 u
Permittivity of a free space (ϵ_o)	$8.85 \times 10^{-12}\ F/m$
Planck's constant, h	$6.6 \times 10^{-34}\ Js$
Proton rest Mass, m_p	$1.67 \times 10^{-27}\ kg$
Radius of the earth, R	$6.4 \times 10^{6}\ m$
Refractive index for glass	1.5
Refractive index for water	1.33
Speed of light in vacuum, c	$3.0 \times 10^{8}\ m/s$
Stefan constant, σ	$5.67 \times 10^{-8}\ W/(m^2k^4)$
Wien Constant, α	$2.9 \times 10^{-3}\ mK$
$(4\pi\ \epsilon_o)^{-1}$	$9.0 \times 10^{9}\ mF^{-1}$
1 atm	$1.105 \times 10^{5}\ Pa$
Rydberg constant	$1.097 \times 10^{7}\ m^{-1}$

SECTION A: MULTIPLE CHOICE QUESTIONS.
Answer ALL Questions.

1. Which of the following phenomena posed a significant challenge to the wave theory of light in the 19th century?
   A. Interference patterns observed in double-slit experiments.
   B. Photoelectric effect observations.
   C. Polarization of light.
   D. Diffraction of light around obstacles.

2. Given two displacements $\vec{A} = (8i + 3j - 2k)$m and $\vec{B} = (4i - 6j + 8k)$m. Determine the magnitude of the displacement $3\vec{A} - \vec{B}$.
   A. 9m
   B. 11m
   C. 15m
   D. 25m

3. From the following options, select the quantity that has the same unit with momentum?
   A. Impulse.
   B. Moment.
   C. Pressure.
   D. Tension.

4. An attempt was made to determine the volume V of a cylinder. The radius r was measured as 3.30 ± 0.05 cm and the length L was measured as 25 ± 0.4 cm. Evaluate the value of V and the uncertainty in this value using the relation: $V = \pi r^2 L$.
   A. (869 ± 0.02) cm^3
   B. (869 ± 0.45) cm^3
   C. (869 ± 4.00) cm^3
   D. (869 ± 40.00) cm^3

5. In the formation of a rainbow, the light from the sun on water droplets undergoes
   A. dispersion only.
   B. only total internal reflection.
   C. dispersion and total internal reflection.
   D. scattering.

6. A student measures the length of a table and finds it to be 1.7 meters. If another student measures the same table and reports the length as 170 centimeters, what can you conclude about their measurements? The measurements are
   A. accurate and precise.
   B. precise but not accurate.
   C. accurate but not precise.
   D. neither accurate nor precise.

7. The electrical resistance from a platinum at 0^0C is 10.50 ohms and 12.80 ohms at 100^0C. Determine the temperature when the resistance is 11.75 ohms assuming that the change in the electrical resistance of the conductor is proportional to the change in temperature.
   A. 50.15 °C
   B. 52.25 °C
   C. 54.35 °C
   D. 64.55 °C

8. A 500 W heater is used to heat 0.6 kg of water from 25^0C to 100^0C in $t_1(s)$. If another 1000 W heater is used to heat 0.2kg of water from 10^0C to 100^0C in $t_2(s)$, determine t_1/t_2.
   A. 2.0
   B. 3.0
   C. 4.0
   D. 5.0

9. Two identical metal rods, one at 50°C and the other at 20°C, are placed in contact with each other. What will happen over time?
   A. Heat will flow from the rod at 20°C to the rod at 50°C until they reach thermal equilibrium.
   B. The rod at 50°C will cool down, and the rod at 20°C will heat up until they reach the same temperature.
   C. Both rods will maintain their initial temperatures as they do not affect each other.
   D. The rod at 20°C will heat up the rod at 50°C until they reach a temperature between 20°C and 50°C.

10. A metal rod of length 200 centimetres at 20°C is heated to 120°C. If the coefficient of linear expansion for the metal is $2 \times 10^{-5}\ °C^{-1}$, what is the final length of the rod?
   A. 0.004 metre
   B. 2.004 metres
   C. 2.040 metres
   D. 4.040 metres

11. The law that relates the emerging electric flux through a closed surface to the enclosed charges is known as
   A. Faraday's law.
   B. Coulomb's law.
   C. Gauss's law.
   D. Ampere's law.

12. When an input step voltage or signal is applied to a circuit, which of the following represents the time response?
   A. Frequency.
   B. Half-life.
   C. Peak period.
   D. Time delay.

13. In a region of space with a uniform electric field, what is the relationship between equipotential surfaces and the electric field lines? Equipotential surfaces

A. are perpendicular to the electric field lines.
B. are parallel to the electric field lines.
C. intersect the electric field lines at a constant angle.
D. and electric field lines have no relationship.

14. A charge of 2 μC is placed at a distance of 4 cm from another charge of 5 μC. What is the electric potential energy of this system, given that the electric constant k is 9×10^9 Nm^2/C^2?

A. 0 J
B. 1.00 J
C. 1.75 J
D. 2.25 J

15. What is the function of the Grid, as part of the electron gun in a cathode ray oscilloscope? It

A. emits electrons when heated by the filaments.
B. accelerates the electrons to a very high speed.
C. controls the brightness of the beam.
D. deflects the beam in the vertical direction.

16. In Bohr's model of the atom, electrons orbit the nucleus at distinct energy levels. Which statement BEST describes the behavior of electrons in these energy levels? Electrons

A. move in random paths within an energy level.
B. can transition between energy levels without absorbing or emitting energy.
C. absorb or emit energy when transitioning between energy levels.
D. remain stationary within an energy level and do not change their positions.

17. In a Compton scattering experiment, an X-ray photon with an initial wavelength of 0.1 nm scatters off an electron at an angle of 60°. Evaluate the change in wavelength of the photon after scattering.

A. 1.21 pm
B. 2.24 pm
C. 4.12 pm
D. 4.48 pm

18. Two forces having magnitude 5 N and 4 N have an angle of 30° between them. What is the scalar product?

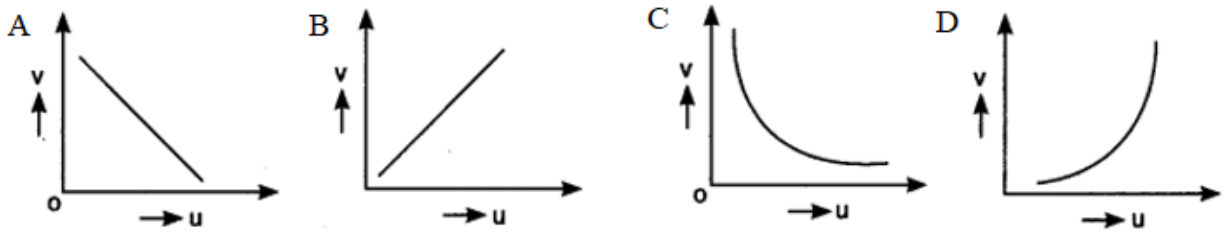
A. $10\sqrt{3}$ N
B. $20/\sqrt{3}$ N
C. $10/\sqrt{3}$ N
D. $5\sqrt{3}$ N

19. An object is thrown horizontally from the open window of a building. If the initial speed of the object is 20 m/s and it hits the ground 2.0 s later, from what height was it thrown?

[Neglect air resistance and assume the ground is level, Take $g = 9.8\ m/s^2$]

A. 4.9 m
B. 9.8 m
C. 10.0 m
D. 19.6 m

20. In an experiment to find focal length of a concave mirror, a graph is drawn between the magnitude of u and v. The graph looks like



21. Heat loss in a transformer due to eddy current can be reduced by
    A. keeping their resistance low.
    B. pouring water on the transformer.
    C. having the iron core laminated.
    D. building air-conditioned house for the transformer.

22. Calculate the magnitude of the force due to a magnetic field of 0.40 T on an electron moving perpendicular to the field with a speed of 4.00 $\times 10^{6}$ ms^{-1}.
    A. 1.28×10^{-13} N
    B. 1.28×10^{-6} N
    C. 2.56×10^{-13} N
    D. 2.56×10^{-6} N

23. Given that a physical quantity has a dimension MLT^{-1}, determine the actual quantity.
    A. Density.
    B. Momentum.
    C. Weight.
    D. Young Modulus.

24. A plane is flying with a velocity of 250 km/h at an angle of 30° north of east. What are the respective horizontal and vertical components of the plane's velocity?

A. 125 km/h, 216.5 km/h

B. 216.5 km/h, 125.0 km/h

C. 250 km/h, 125.0 km/h

D. 125.0 km/h, 250 km/h

25. If the current in a solenoid (coil of wire) increases uniformly from 0 A to 3 A in 3×10^{-3}s, calculate the magnitude of the self induced e.m.f. (Take the inductance as 25 mH).

A. 10 V

B. 18 V

C. 25 V

D. 31 V

26. Two electrical appliances, A and B, are both rated at 1200 W. However, appliance A operates at 120 V, and appliance B operates at 240 V. Compare the current flowing through each appliance.

A. Appliances A and B have the same current.

B. The current cannot be determined from the given information.

C. Appliance B has a higher current than appliance A.

D. Appliance A has a higher current than appliance B.

27. Which of the following is respectively arranged as dimension of Impulse, momentum and work?

A. ML^2T^{-1}, MLT^{-1} & ML^2T^{-2}

B. MLT^{-1}, MLT^{-1} & ML^2T^{-2}

C. ML^2T^{-2}, MLT^{-1} & MLT^{-1}

D. MLT^{-1}, ML^2T^{-2} & MLT^{-1}

28. How much heat must be added to 50 kg of lead at 30 °C to cause it to melt? Specific heat capacity of lead is 130 $Jkg^{-1}C^{-1}$, it melts at 330 °C and its heat of fusion is 24600 Jkg^{-1}.
A. 1950000 J
B. 123000 J
C. 1827000 J
D. 3180000 J

29. Calculate the force on a charge +6 μC moving with a velocity 2×10^6 ms^{-1} in a field of magnetic flux density 0.45 T parallel to the field.
A. 5.4N
B. 3.4N
C. 0.0N
D. 3.0N

30. A beam of X-rays with an intensity of 500 W/m^2 is incident on a lead sheet with a thickness of 2 cm. If the linear attenuation coefficient of lead for X-rays is 0.1 cm^{-1}, compute the intensity of the X-ray beam after passing through the lead sheet.
A. 400 W/m^2
B. 409 W/m^2
C. 449 W/m^2
D. 499 W/m^2

31. Choose from the given options, the correct property of X-rays. They
A. affect photographic plates.
B. are not diffracted by crystals.
C. are deflected by electric or magnetic fields.
D. have very low frequency.

32. Which of the following types of motion describes the motion of smoke particles from a chimney?
A. Rotational motion.
B. Circular motion.
C. Random motion.
D. Oscillatory motion.

33. When the period of the satellite in its orbit equals the period of rotation of the earth about its axis, the orbit is called
A. inertial orbit.
B. terminal orbit.
C. parking orbit.
D. sinusoidal orbit.

34. A boat is moving northward with a velocity of 18 m/s, and a crosswind is blowing from the east with a velocity of 15 m/s. Determine the resultant velocity of the boat.
A. 33.0 m/s north
B. 23.4 m/s north-east
C. 17.2 m/s east
D. 3.0 m/s north-east

35. The random distribution of the kinetic and potential energies associated with the particles of a system is known as
A. Internal energy
B. Thermal equilibrium
C. Carnot engine
D. Viscosity

36. Which of the following statements is NOT true about the resistivity of a metal? Resistivity

A. is the reciprocal of conductivity.

B. varies inversely with temperature.

C. varies linearly with temperature.

D. is proportional to resistance.

37. Which of the following options BEST describes a “Blackbody”?

A. A perfect absorber and emitter of all electromagnetic radiation.

B. An object that appears completely black and absorbs all visible light.

C. A material with low thermal conductivity.

D. A substance that reflects all incident light.

38. A capacitor is connected to a 12-volt battery and stores a charge of 4 μC. What is the energy stored in the capacitor?

A. 12 μJ

B. 24 μJ

C. 48 μJ

D. 96 μJ

39. Moseley's Law finds application in which of the following?

A. Determining the mass of subatomic particles.

B. Identifying elements in crystal structures.

C. Understanding the behaviour of plasma in fusion reactors.

D. Calculating the energy levels of electrons in atoms.

40. The Bohr radius is defined as

A. $r_n = \frac{\varepsilon_0 h^2}{mZe^2} . n^3$

B. $r_n = \frac{\varepsilon_0 h^2}{\pi mZe^2} . n^2$

C. $r_n = \frac{\varepsilon_0 h^2}{\pi mZe^6} . n^2$

D. $r_n = \frac{h^9}{\pi mZe^2} . n^2$

41. Which of the following properties of X-rays allows them to penetrate matter and produce images in medical imaging techniques?

A. X-rays travel at the speed of light.

B. X-rays have wavelengths longer than visible light.

C. X-rays have high energy and short wavelengths.

D. X-rays have a higher frequency compared to gamma rays.

42. A spring of force constant k is stretched a certain distance. It takes twice as much work to stretch a second spring by half this distance. The force constant of the second spring is?

A. k

B. 2 k

C. 4 k

D. 8 k

43. A sample of gas is initially at a temperature of 300 K and a pressure of 2 atm. If the temperature is doubled while the volume is kept constant, what will be the new pressure of the gas if it behaves ideally?

A. 1 atm

B. 2 atm

C. 4 atm

D. 8 atm

44. An electric current of 3 A flowing through an electric heating element of resistance 20 Ω embedded in 1000 g of an oil raises the temperature of the oil by 10 ^{o}C in 10 s. What is the specific heat capacity of the liquid?

A. 0.18Jg^{-1} ^{o}C^{-1}
B. 2.18Jg^{-1} ^{o}C^{-1}
C. 0.28Jg^{-1} ^{o}C^{-1}
D. 0.08Jg^{-1} ^{o}C^{-1}

45. What principle underlies the operation of a CT (Computed Tomography) scanner?

A. Doppler effect in sound waves.
B. Refraction of X-rays through a medium.
C. Interference of light waves from multiple X-ray projections.
D. Reconstruction of 3D images from multiple X-ray projections.

46. In placing your right palm in front of a plane mirror, its image appears in the mirror as a left palm. This is referred to as

A. angular inversion.
B. lateral inversion.
C. angular magnification.
D. lateral magnification

47. Which of the following illustrates the definition of angle dip? The angle between the

A. direction of the magnetic North and the direction of geographic North.
B. direction of the earth's magnetic flux and the horizontal.
C. earth's magnetic field and the vertical.
D. meridian and the magnetic north pole.

48. Compute the accelerating potential if the frequency of an emitted X-ray is $1.6 \times 10^{16} Hz$ $(e = 1.6 \times 10^{-19} C, \quad h = 6.63 \times 10^{-34} Js)$.

A. 11.4V

B. 44.7V

C. 66.3V

D. 88.0V

49. In a medical procedure, a radioactive isotope with a half-life of 6 hours is injected into a patient. If the initial activity was 80 mCi, what will be the activity after 18 hours?

A. 5 mCi

B. 10 mCi

C. 20 mCi

D. 40 mCi

50. The wavelength of a sound wave in a certain material is 36.0 cm. What is the speed of the sound wave in a period of 1.1 millisecond?

A. 300. m/s

B. 327.3 m/s

C. 342 m/s

D. 360 m/s

SECTION B: PHYSICS ESSAY QUESTIONS
Answer FOUR Questions in ALL; ONE from each Course.

PHYSICS 001: MECHANICS AND PROPERTIES OF MATTER

1. (a) (i) A vernier caliper has 20 divisions on the vernier scale. 1cm on the main scale is divided into 10 equal parts. Determine the least count on the vernier scale. **[2 Marks]**

(ii) Complete the below table: **[2 Marks]**

S/N	Physical quantity	Unit
I		Js^{-1}
II	Surface tension	
III		Nm
IV	Electric charge	

(b) A mason working at a construction site drops his trowel from a height of 125 m. How long does it take to reach the ground? [Take g = 10 ms^{-2}] **[2 Marks]**

(c) (i) A body of mass of 4×10^{-3} kg is placed on top of a spiral spring of force constant 400 Nm^{-1}. If the spring is compressed downwards by 0.02 m and then returns, calculate the height to which the body is projected. [g = 10 ms^{-2}] **[2 Marks]**

(ii) The engine of a train produces a force of 5000 N when moving at 50 m/s. What is the power rating of the engine? **[2 Marks]**

2. (a) Explain the following:

(i) Centre of gravity;

(ii) Angular momentum;

(iii) Inertia.

(iv) Terminal speed.

[2 Marks]

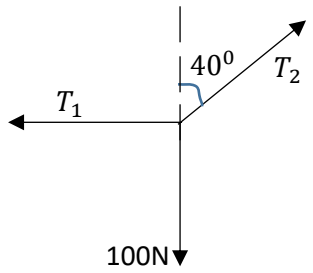
(b) (i) State Newton's second law of motion.

[1 Mark]

(ii) If $\vec{A} = -5i + 6j - 3k$ and $\vec{B} = 7i + 8j + 4k$, find $\vec{A} \times \vec{B}$.

[3 Marks]

(c) A body of mass 10 kg is kept in equilibrium by two ropes as shown in the diagram below. If one rope pulls the body in a horizontal direction and the other in a direction 40^0 to the vertical, calculate the tension in each rope. **[4 Marks]**



PHY 002: HEAT, WAVES AND OPTICS

3. (a) (i) State any **FOUR** components of the electromagnetic spectrum in order of their decreasing wavelength. **[2 Marks]**

(ii) State **TWO** properties of electromagnetic waves that distinguish them from mechanical waves. **[1 Mark]**

(b) (i) State and define the **THREE** modes of heat transfer.

[3 Marks]

(ii) A filament of a light bulb of surface area $7 \times 10^{-6}\ m^2$ operates at 1727 °C. Calculate the electrical power which must be supplied to filament if its emissivity is 0.91.

[Stefan constant = $5.67 \times 10^{-8}\ Wm^{-2}K^{-4}$] **[2 Marks]**

(c) A metal rod of length 80cm and radius 3.5 cm attains a steady state with the end exposed to a temperature of 200 ^{o}C and the other end at 40 ^{o}C. Determine the time rate of heat transfer. [Thermal conductivity of the metal is 210 $Wm^{-1}\,{}^{o}C^{-1}$, $\pi = 3.14$] **[2 Marks]**

4. (a) (i) Distinguish between heat and temperature. **[2 Marks]**

(ii) State any **THREE** reasons why mercury is preferred to alcohol as a thermometric liquid. **[3 Marks]**

(b) (i) State Boyle's law and express its mathematical relations. **[2 Marks]**

(ii) A gas contained in a cylinder is of volume $1.0 \times 10^{-2}\ m^3$ at temperature 325 K and pressure $2.7 \times 10^{5}\ Nm^{-2}$. When some of the gas is used at constant temperature, the pressure falls to $1.2 \times 10^{5}\ Nm^{-2}$. Calculate the number of moles of the gas used.

[Take $R = 8.3\ Jmol^{-1}K^{-1}$] **[3 Marks]**

PHY 003: ELECTRICITY AND MAGNETISM

5. (a) Draw a diagram to illustrate the magnetic field between the south poles of two bar magnets. On your diagram, indicate the "neutral point" with K. **[2 Marks]**

(b) A lamp used in a JUPEB Foundation School is rated 60 W, 220 V with a 4400 V A.C supply and operated by a transformer which is 95% efficient. Calculate the

(i) ratio of the number of turns between the primary and secondary coils of the transformer respectively.

(ii) current supplied by the mains circuit. **[4 Marks]**

(c) (i) In an A.C circuit, the peak value of the potential difference is 180 V. What is the instantaneous potential difference when it has reached two-third of a cycle? **[2 Marks]**

(ii) State **ONE** difference between the reactance and the impedance of an A.C circuit. **[1 Mark]**

6. (a) (i) Explain the term "Temperature coefficient of resistance" **[1 Mark]**

(ii) The resistance of a conductor is 3 kΩ at a temperature of 150 ^{o}C**,** if the temperature coefficient of resistance is 0.005 $^{o}C^{-1}$, at what temperature will the resistance become 5 kΩ. **[2 Marks]**

(b) (i) State the Flemings Right hand rule. **[2 Marks]**

(ii) A solid of 2000 turns coil is 75 cm long and carries a current of 2 A, calculate magnetic flux density at the centre of the solenoid. Hence determine the force exerted on an electron moving with a velocity 3000 m/s through the interior of the solenoid at an angle of 30° relative to the solenoid axis. **[3 Marks]**

(c) A 10 μF capacitor is connected in series with a 0.4 MΩ resistor. The DC voltage supply is 200 V, calculate the voltage in the capacitor, 5 s after the power is supplied. **[2 Marks]**

PHY 004: Modern Physics

7. (a) An electron makes a transition from -3.4 eV energy level to -13.6 eV energy level. Calculate the:

(i) amount of energy in joules lost as a result of the transition.

(ii) frequency of the emitted radiation. [Take h = 6.63×10^{-34} Js] **[3 Marks]**

(b) In a thermonuclear reaction, the difference between the initial and final mass is 1.0×10^{-29} kg. What is the amount of energy released in the process? **[2 Marks]**

(c) A photoelectric surface has a threshold frequency of 4.02×10^{14} Hz. If light of frequency 5.00×10^{15} Hz illuminates the surface, calculate the:

(i) work function;

(ii) threshold wavelength;

(iii) kinetic energy of the emitted photons. [h = 6.63×10^{-34} Js]

[5 Marks]

8. (a) Explain the term threshold frequency and work function as used in the concept of photo electric effect . **[3 Marks]**

(b) Consider the popular nuclear reaction that uses deuterium as fuel;

$$3({}^{2}_{1}H) \longrightarrow {}^{4}_{2}He + {}^{1}_{1}H + {}^{1}_{0}n$$

(i) Determine the mass defect of a single reaction given the following data: $({}^{2}_{1}H) = 2.014u, {}^{4}_{2}He = 4.0026u, {}^{1}_{1}H = 1.0078u, {}^{1}_{0}n = 1.0087u$

(ii) Determine the energy in Joules released during a single fusion reaction. **[4 Marks]**

(c) A clinical X-ray tube was designed to operate at 24.8 kV. Determine the maximum possible wavelength of the radiation emitted from the tube**.** **[3 Marks]**